Partner-Specific Affective Precision in Social Active Inference

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Abstract. We model affect in social settings as metacognition: an estimate of whether the focal agent's current partner model warrants confident action, rather than a direct signal of what another agent will do or a record of reward and punishment. This distinction matters in multi-partner environments, where an agent may have a reliable model of one partner, an uncertain model of another, and a predictably hostile model of a third. We formalize this idea in active inference by extending affective precision to relationship-specific model-fitness estimates. Each partner receives its own β estimate, updated only from that partner's behavioral evidence, which modulates policy precision when acting toward that partner.
Simulations in a multi-partner trust game show that partner-local affective precision tracks predictability rather than value: a reliably defecting and a reliably cooperative partner can both support high precision, because what they share is predictability, not positive value. In decision-open regimes, deployment changes arise through action sharpening rather than improved belief quality; in partner-choice settings, precision modulation produces modest partner-choice redistribution without a clear aggregate payoff separation. Under abrupt behavioral change, relationship-specific confidence reshapes action deployment through policy precision; payoff effects are modest and uncertain, and the temporal benefit of accumulated confidence depends on whether the social portfolio permits rapid reallocation away from the changed partner. Precision gain and priors over model fitness shape computational profiles of social trust calibration. These results describe affective precision as a relationship-specific signal that regulates confidence in social action.

Keywords: Active inference · Affective precision · Social metacognition · Multi-agent systems · Trust · Individual differences

1 Introduction

Social decision-making involves three problems that are useful to separate computationally. The first is inferring what a partner is like—whether they tend to cooperate, defect, or behave unpredictably. The second is modeling what they know or intend—the theory-of-mind question of how another agent represents the world and plans action [10, 11, 17]. The third is estimating whether one's *current* partner model is reliable enough to guide confident action. Accurate type inference and sophisticated mentalizing do not by themselves solve this third problem. Knowing that someone is unreliable is not the same as knowing how cautiously to treat them; knowing that someone is predictable is not the same as knowing how strongly to commit to that prediction. Many formal models specify what an agent infers about others, while treating the confidence placed in those inferences as a fixed or global quantity.

We model social affect as a metacognitive estimate of whether the focal agent's current partner model warrants confident action, rather than as a direct readout of another agent's future behavior or a record of reward and punishment. In psychology and neuroscience, affect is often described in terms of valence and arousal—the felt pull toward or away from situations, relationships, and social commitments [2]. Functionally, however, it can regulate how strongly beliefs are expressed in action: affect assigns action-relevant weight to what an agent already infers, shaping the degree of commitment rather than the content of inference. An agent that infers accurately but commits poorly wastes its social knowledge; one that commits confidently on fragile models exposes itself to exploitation.

The active inference framework [7, 8, 16] provides a formal route from this observation to a computational mechanism. Agents minimize variational free energy to update beliefs and evaluate candidate policies by their expected free energy [9, 6]. Action selection is precision-weighted: the same beliefs can produce decisive or tentative behavior depending on γ, the parameter controlling how sharply the agent commits to its preferred policies. Hesp et al. [13] made affect endogenous by modeling it as active inference over a higher-order belief β representing subjective model fitness. An affective charge signal ϕ drives updates to β, rising when the model performs better than a prior baseline and falling when it performs worse. On this view, affect is metacognitive rather than mentalistic: it tracks model fitness rather than partner intentions or reward history alone.

Multi-partner settings require this confidence estimate to be relationship-specific. Model fitness can differ sharply across partners: a person can carry a well-validated model of a longtime collaborator, a fragile model of a new acquaintance, and an actively revising model of someone who has recently behaved unexpectedly—and these confidence states should not be collapsed into a single shared precision budget. We therefore assign each relationship its own β estimate, updated from that partner's behavioral evidence and used to set action confidence toward that partner.

We evaluate the mechanism in a repeated multi-partner trust game across binary and graded decision regimes, partner-choice conditions, abrupt betrayal, and profile-style parameter perturbations. The results characterize both the functional role and the temporal boundary conditions of partner-local social metacognition. The simulations evaluate three predictions: partner-local affective precision should track model reliability more than realized payoff; its behavioral effects should arise through action confidence rather than improved partner-state beliefs; and precision gain together with prior model fitness should shape temporally structured profiles of social trust revision.

2 Methods

A focal active-inference agent plays a repeated trust game with four partners whose types and stances are hidden and must be inferred from behavioral evidence. The agent maintains a separate generative model for each partner, allowing evidence and confidence to remain relationship-specific. Partner-local affective precision is layered on top of these models: each partner has its own model-fitness estimate, which modulates how confidently the agent acts on that partner's inferred state.

2.1 Trust game

The trust game [4, 14] is a natural testbed because partner types are hidden and must be inferred, creating a genuine gap between social knowledge and actionable confidence. The policy space can also be varied: binary settings offer two actions, graded settings introduce several investment levels, and partner-choice regimes make approach and avoidance explicit by requiring the agent to select whom to engage.

Each partner has a latent *type* (cooperator, reciprocator, exploiter, or random) and a latent *stance* toward the agent (trusting, neutral, or hostile)—neither directly observable; both must be inferred from behavioral evidence. Episodes run for up to 200 rounds.

In the binary game both players choose to cooperate or defect. Table 1 gives the focal agent's payoffs.

Table 1: Focal agent payoffs in the binary trust game.

	Partner cooperates	Partner defects
Agent cooperates	3	-1
Agent defects	5	1

The structure poses a social inference problem: defection is locally tempting, but cooperative partners are worth identifying and maintaining, so the agent must infer type and stance to act well over repeated interaction. In the graded regime, the binary choice is replaced by a discretized investment level; the agent commits a resource along a continuous scale rather than making a binary commitment. In the partner-choice regime, the agent selects which partner to engage before choosing an action each round.

2.2 Per-partner generative models

Given this environment, the focal agent maintains a separate discrete generative model $\mathcal{M}_k$ for each partner k, rather than a single pooled social model. This factorization keeps evidence about each partner independent and allows action confidence to vary across relationships. Each model encodes observation likelihoods, transition dynamics, preferences, priors, and policy priors using the standard $A/B/C/D/E$ structure of discrete active inference. Payoff is represented as an observation modality rather than as an external reward cache, so preferences over outcomes remain part of the agent's generative model. State and policy inference use the `pymdp` package for discrete active inference [12, 6].

2.3 Generative model specification

Each partner model contains three latent factors: partner type $s_k^{\text{type}} \in \{\text{cooperator}, \text{reciprocator}, \text{exploiter}, \text{ran}$ partner stance $s_k^{\text{stance}} \in \{\text{trusting}, \text{neutral}, \text{hostile}\}$, and a deterministic own-action bookkeeping factor s^{own}. The own-action factor is not a psychological hidden variable; it makes the focal agent's executed action available to the payoff likelihood in a standard factorized POMDP representation. Observations are partner action $o_k^{\text{act}} \in \{\text{cooperate}, \text{defect}\}$ and focal payoff $o_k^{\text{pay}} \in \{-1, 1, 3, 5\}$.

The partner-action likelihood A_k^{act} is defined by type- and stance-conditioned cooperation probabilities. This table operationalizes predictive reliability: both reliably cooperative and reliably defecting partners can be predictable, even though their payoff consequences differ. Payoff enters as an observation modality $P(o_k^{\text{pay}} \mid s^{\text{own}}, s_k^{\text{type}}, s_k^{\text{stance}})$, with partner action marginalized through the cooperation likelihood. Thus payoff is deterministic in the environment but represented as a likelihood in the focal agent's generative model. Stance transitions are action-dependent: cooperation tends to build trust, defection tends to damage trust, and recovery from hostility is slower than collapse. Full $A/B/C/D/E$ matrices, cooperation probabilities, stance-transition matrices, priors, and payoff construction are provided in Appendix A.

2.4 Partner-local affective precision

Hesp et al. [13] treat policy precision as an inferred quantity rather than a fixed hyperparameter. The agent maintains a higher-order belief β over inverse policy precision, representing uncertainty about subjective model fitness: how reliably the generative model is supporting confident action. An affective charge signal ϕ drives updates to β, shifting the agent toward higher confidence when the model performs better than a prior baseline and toward lower confidence when it performs worse.

We extend this to the per-partner case by giving each partner its own β_k estimate, updated from that partner's behavioral evidence alone. After interacting with partner k, the agent computes surprisal from the predicted probability of the observed partner response:

$$\epsilon_k = -\log P\left(o_k^{\text{act}} = \hat{o}_k^{\text{act}} \;\middle|\; q(s_k^{\text{type}}, s_k^{\text{stance}})\right). \tag{1}$$

This is computed from partner *action* rather than payoff because action is the most direct test of type-by-stance model fit: a reliably defecting partner produces no action surprise even though the payoff is poor. We set the neutral fitness baseline at $\sigma_0^2 = (-\log 0.5)^2$, so a fifty-fifty binary prediction carries zero affective charge. Affective charge is then:

$$\phi_k = \alpha\left(\sigma_0^2 - \epsilon_k^2\right), \tag{2}$$

where α is a gain parameter. Positive ϕ_k means partner k's model predicted better than the baseline; negative ϕ_k means it predicted worse. The categorical posterior $q(\beta_k)$ is updated by combining a slowly evolving persistence prior with a charge-based pseudo-likelihood that shifts mass toward low β after accurate prediction and toward high β after surprise; the full update rule is given in Appendix B.

The agent then uses the posterior mean to set policy precision:

$$\gamma_k = \gamma_{\text{base}} \,/\, \mathbb{E}_{q(\beta_k)}[\beta_k]. \tag{3}$$

Here, β_k represents inverse precision or model-fitness uncertainty: higher β_k means the model has been less reliable, producing a broader, less committed policy distribution; it is the inverse relationship that drives action sharpening. The mechanism thus translates accumulated evidence about predictive reliability into action confidence—independently for each social relationship.

The β_k posterior is therefore treated as an auxiliary precision tracker rather than as a hidden state within the generative model; the implications of this design choice are discussed in Section 4.

Cross-partner policy selection. In partner-choice regimes, policies specify both which partner to engage and how to act toward that partner. Each candidate policy branch inherits the precision of the partner it targets. Precision is applied to centered branch logits so that γ_k sharpens or flattens action selection within a partner branch without changing that branch's mean score in cross-partner comparison. This allows partner-local confidence to affect deployment while avoiding an artifact in which low precision makes uniformly poor partner branches spuriously attractive. Full details are provided in Appendix B.

Partner implementation. Reported experiments use one focal active-inference agent interacting with four environment-side partners. Partners maintain latent type and stance, sample actions from the type-by-stance cooperation probabilities, and undergo action-dependent stance transitions, but they do not perform variational inference or maintain their own affective precision estimates. This isolates the focal agent's partner-local precision mechanism before extending the model to reciprocal active-inference partners. Full implementation details are provided in Appendix A.

Simulation scale. Unless otherwise stated, simulations use 200-round episodes. Profile-scale and mixed-volatility experiments use 20 seeds per condition, while central behavioral comparisons use 30 seeds per condition. The abrupt-betrayal condition uses 120 rounds because the switch is scheduled explicitly and the main readouts are post-switch behavior. Full condition definitions, schedules, metrics, and seed counts are provided in Appendix C.

Code availability. Simulation code, analysis scripts, and experiment configurations will be released in a public repository upon acceptance.

3 Results

The simulations evaluate three aspects of relationship-specific confidence. First, partner-local affective precision should track predictive reliability more than realized value, and pooling evidence across partners should weaken this relationship-specific signal. Second, its behavioral effects should arise through action sharpening rather than improved partner-state beliefs, with consequences for both policy entropy and partner engagement. Third, under social change, precision dynamics should reveal how gain and prior model fitness shape trust revision. The subsections below evaluate these questions using reliability probes, shared-β ablations, deployment controls, partner-choice regimes, abrupt betrayal, and profile-style perturbations.

3.1 Precision tracks predictability, not partner value

The affective charge formula uses action surprisal rather than payoff, so β_k is sensitive to behavioral consistency rather than realized value. Once the agent has inferred a partner's type and stance, a reliably defecting partner can generate low surprisal despite poor payoff, while an erratic partner can generate high surprisal despite occasionally favorable outcomes. The precision signal is therefore expected to track the quality of the agent's predictive model, not the desirability of the partner's behavior.

The 30-seed model-fitness analysis separated reliability from value. Across partner-seed units, partner-local β_k shows a stronger absolute association between precision and action surprisal than between precision and realized payoff (0.949 versus 0.748). Because payoff, surprise, and exposure are correlated in partner-choice episodes, we also computed an active-encounter partial readout: after controlling encounter imbalance, the partner-local precision–surprise association remains strong (0.882), whereas the precision–payoff association falls to 0.130. A single shared β tracker preserves the ordering but with weaker relational specificity (partial associations 0.380 versus 0.049). Aggregate payoff followed a different pattern (partner-local: 483.5; shared: 517.6; no-affect: 542.1), separating the precision signal from reward maximization. Partner-local precision tracked predictive model fitness more closely than realized partner value. Figure 1 visualizes this dissociation for partner-local versus shared precision tracking.

3.2 Shared precision tests relational specificity

The locality probe tests whether confidence remains tied to the partner whose evidence produced it. Pooling evidence across partners weakens this relational structure. A shared β tracker preserves the coarse ordering between predictive surprise and realized payoff, but attenuates the partner-specific dissociation

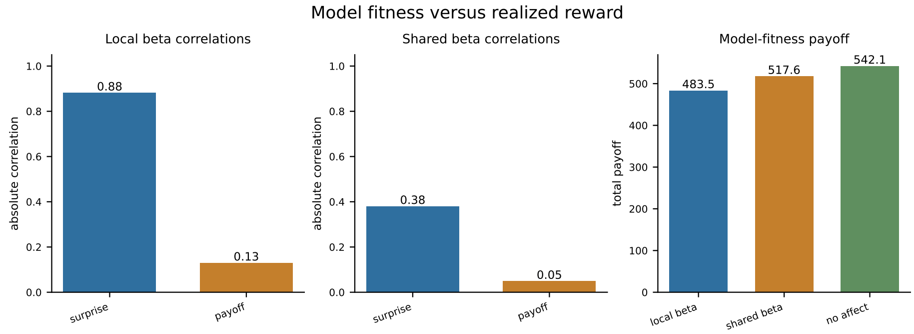

Fig. 1: Partner-local precision tracks action surprisal more strongly than realized payoff. The panel contrasts active-encounter partial associations between mean partner precision and surprise versus payoff for partner-local β_k and a shared β tracker, plus aggregate payoff. Partner-local tracking preserves the surprise-over-reward dissociation; pooling evidence across partners attenuates relational specificity.

observed under partner-local precision. In the active-encounter partial readout, partner-local precision shows a stronger association with surprise than payoff, whereas shared precision compresses both associations toward weaker values.

This comparison isolates the locality of the precision signal. Pooling evidence blurred model-fitness estimates across relationships, while partner-local β_k preserved the link between each partner's evidence and that partner's action confidence. The mixed-volatility analyses below examine a related issue: how stronger precision dynamics can increase payoff while also increasing false-positive confidence reduction toward stable partners. A mixed-volatility shared-versus-local comparison would test the behavioral consequences of this attenuation. The present analysis shows that relationship-specific precision preserves the link between each partner's evidence and that partner's confidence signal.

3.3 Behavioral effects arise through action sharpening

A tracked-only ablation updates $q(\beta_k)$ normally but fixes $\gamma_k = \gamma_{\text{base}}$, allowing the precision estimate to be analyzed without letting it affect action selection. In the open graded decision regime, full partner-local precision lowers policy entropy relative to both baselines (8.6 versus 8.8 for no-affect and tracked-only), while total payoff is flat-to-slightly lower (1851.3 versus 1864.2 for both baselines). Tracked-only remains near the no-affect baseline, while full precision modulation lowers entropy. The behavioral effect therefore enters through the $\beta_k \to \gamma_k$ deployment pathway rather than through improved partner-state inference. Figure 2 summarizes the three-way contrast across affect, no-affect, and tracked-only variants.

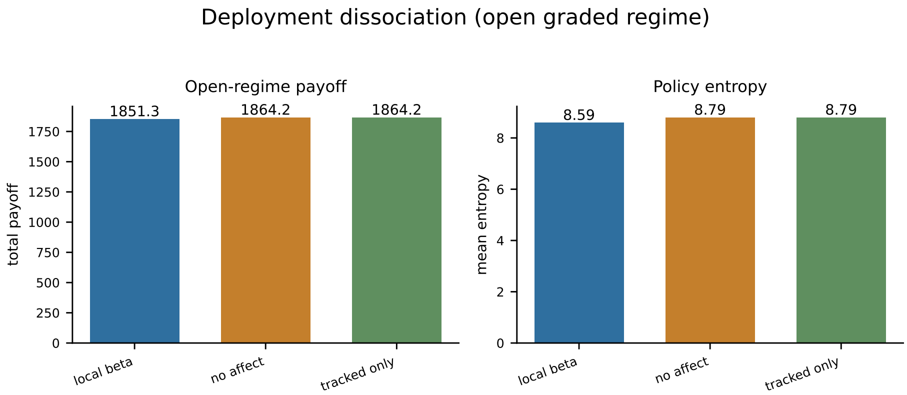

Fig. 2: Open graded regime: partner-local affective precision lowers policy entropy relative to no-affect and tracked-only controls without a corresponding payoff advantage. Tracked-only updates β_k but fixes γ_k; its deployment readouts match the no-affect baseline, isolating the entropy effect to the $\beta_k \to \gamma_k$ action-confidence pathway.

3.4 Partner choice shifts under precision modulation

In partner-choice regimes, precision modulation can affect both action selection and partner selection. Policy entropy under precision modulation is 4.66 versus 4.81 without it, and engagement shows a modest redistribution across partner types. The partner-local agent selected the cooperator on 36.6% of rounds and the reciprocator on 31.6%, while allocating 13.8% and 18.0% of choices to the exploiter and random partners, respectively. The no-affect control selected the same partners at 34.8%, 31.8%, 16.2%, and 17.2%, respectively. Aggregate payoff remains similar at this scale (377.3 versus 385.3), consistent with selection reorganization occurring before cumulative payoff differences have time to accumulate.

These allocation readouts show modest partner-choice redistribution without a clear payoff separation at this replication scale.

3.5 Abrupt betrayal tests confidence under change

When a previously cooperative partner changes stance mid-episode, the agent enters the post-switch period with confidence accumulated under a model that no longer matches the partner's behavior. In the 30-seed abrupt-betrayal analysis, partner-local affective precision produces a small payoff advantage relative to the no-affect control (1185.9 versus 1172.1), but the paired bootstrap interval for the difference crosses zero (-25.2 to 53.2). The clearest effect is lower policy entropy under precision modulation (8.36 versus 8.74; bootstrap interval for the difference -0.62 to -0.14), confirming that the action-confidence pathway is active.

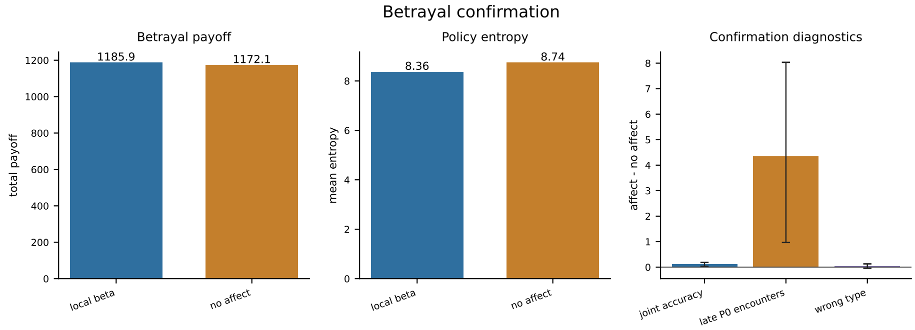

Fig. 3: Abrupt betrayal in the partner-choice regime (30 seeds). Partner-local affective precision lowers policy entropy relative to the no-affect control; the payoff difference is positive but small, and its bootstrap interval crosses zero. Joint partner-type accuracy is also higher under precision modulation, indicating deployment change with modest behavioral benefit.

Joint partner-type accuracy is higher under precision modulation (0.372 versus 0.266; bootstrap interval for the difference 0.034 to 0.185). Under abrupt social change, partner-local precision produced lower entropy, higher joint partner-type accuracy, and an uncertain payoff advantage. The benefit of accumulated confidence depended on portfolio structure and the speed with which the agent could redirect commitment after change.

The parameter analyses below vary precision gain and prior model fitness to examine how this confidence-revision process changes across profiles.

3.6 Precision dynamics as computational profiles of social trust calibration

We varied precision gain α and the prior over model fitness to examine how confidence-revision dynamics change across social contexts. These parameter settings provide computational analogues for empirically studied variation in social trust calibration [3, 1, 5], but they are not fitted to human behavior.

Initial parameter perturbations (Appendix D) show that flattened precision dynamics can reduce action commitment without improving payoff: the low-gain variant compresses β_k range to 0.24 and raises entropy to 8.9, near the no-affect and tracked-only baselines, whereas expanded-dynamics variants widen β_k movement without monotonic payoff gains. We therefore sweep gain and prior structure directly rather than treating trust calibration as a single high-versus-low axis.

Precision-gain axis. Sweeping α across $\{0.05, 0.1, 0.3, 0.5, 1.0, 2.0, 4.0, 8.0\}$ shows that β_k movement range increases monotonically with α in the betrayal regime. Low-α agents ($\alpha \leq 0.1$) show near-zero exploitation rates and Gini coefficients of

0.423 and 0.397, producing diffuse, slowly-updating engagement patterns. High-α agents ($\alpha \geq 4.0$) show Gini coefficients of 0.366 and 0.408 with recovery times of 28.8 and 13.0 rounds respectively—faster detection of behavioral change in some runs, but higher variance in recovery outcomes. Payoff does not rank α values monotonically across either regime (betrayal: range 1914.6–1986.2; open graded: 1906.3–1987.7), indicating that the behavioral consequence of precision-gain depends on task context rather than gain magnitude alone. Extended alpha-sweep diagnostics are in Appendix D.

Gain-prior profiles. Crossing prior belief (naive versus cautious) with α (low versus high) defines four behavioral profiles tested in the betrayal environment (20 seeds). Low-α profiles (naive-stubborn and avoidant-rigid) show substantially smaller β_k range (0.32–0.36) than high-α profiles (0.89–1.08), confirming that gain controls the magnitude of precision dynamics. Naive-stubborn shows the lowest Gini (0.304), reflecting maximally diffuse partner selection consistent with a slowly-updating confidence profile. Avoidant-rigid shows the lowest payoff (1889), as the cautious prior prevents confidence from building even toward reliably cooperative partners. The anxious-reactive profile shows the highest post-betrayal commitment errors toward the switched partner (post-betrayal P0 selection: 0.319; high investment rate: 0.121), reflecting rapid confidence accumulation that becomes a liability when a partner's behavior changes. The default agent achieves the highest payoff among these variants (1991), suggesting that payoff depends on calibration within the gain-prior space. Quadrant figures and supporting diagnostics are in Appendix D.

Forgiveness and trust repair. In the forgiveness paradigm, the betrayed partner reverts to cooperative/trusting behavior after round 120. The main pattern separates reengagement from restored confidence. No-affect and cautious-low-α agents reengage most often after repair (selection rates 0.593 and 0.630), while default and naive-high-α agents reengage less (0.475 and 0.560). Payoff recovery is near baseline across all conditions (0.996–1.033), separating reengagement from payoff recovery rather than producing a simple "more affect means better forgiveness" pattern. Low gain produces muted β_k dynamics and rapid reapproach in this task (cautious-low latency 1.5 rounds; naive-low 4.6), whereas high-gain profiles show larger confidence swings but not a higher payoff-recovery ceiling. Trust repair separates willingness to sample the reverted partner from restoration of model-fitness confidence.

Mixed-volatility discrimination. The mixed-volatility environment places four partners with heterogeneous volatility in the same episode: a stationary cooperator (P0), a stationary exploiter (P1), an abrupt-switch partner (P2), and a gradual-drift partner (P3). Under partner choice, the agent must maintain stable engagement with P0 while updating away from P2 and P3 after their behavioral changes.

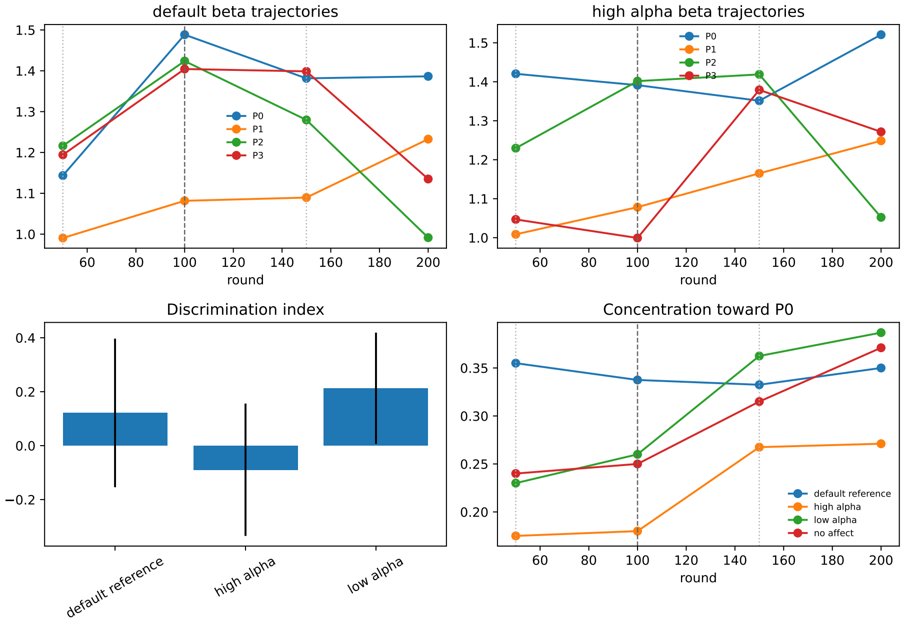

Fig. 4: Mixed-volatility partner-choice environment (20 seeds). Four partners differ in volatility: stationary cooperator (P0), stationary exploiter (P1), abrupt switch (P2), and gradual drift (P3). Higher precision-gain α can raise payoff while worsening discrimination and increasing false-positive confidence reduction toward P0.

Table 2: Mixed-volatility metrics by precision profile (20 seeds).

Variant	Payoff	Discrimination	P0 false positive	β range
default	1976	0.121	0.422	0.933
high-α	2054	-0.089	0.484	1.049
low-α	1927	0.213	0.359	0.182
no-affect	1934	–	0.273	–

The false-positive rate—unwarranted confidence reduction toward the stationary P0—increases with α: high-α (0.484), default (0.422), low-α (0.359), no-affect (0.273). The discrimination index does not simply improve with gain: low-α gives the strongest discrimination score (0.213), while high-α is negative (-0.089) despite achieving the highest payoff. Figure 4 shows that payoff did not increase monotonically with gain. In this environment, high gain amplified reactivity and false positives toward stable partners, while payoff depended on how that reactivity interacted with partner allocation.

Together, the profile experiments show a family of confidence-revision profiles determined by gain and prior initial conditions. Across environments, confidence dynamics, partner allocation, and payoff dissociated rather than following a monotonic payoff ranking. Extended profile tables, alpha-sweep diagnostics, and forgiveness/trust-repair analyses are provided in Appendix D.

4 Discussion

We introduced a partner-local affective precision mechanism for social active inference. Each partner model maintains its own estimate of model fitness, updated from that partner's behavioral evidence, and uses this estimate to modulate policy precision during action selection. Across experiments, this mechanism separated social prediction from social commitment: the agent could infer what a partner was likely to do while separately estimating how confidently to act on that inference.

First, the model separated reliability from value. Partner-local precision tracked action surprise more strongly than realized payoff, even when aggregate reward favored the non-affective baseline. This followed from the update rule: β_k changed with the predictability of the partner's behavior rather than with whether the partner was beneficial. A reliably cooperative partner and a reliably defecting partner could therefore both support high precision. Affective precision therefore behaved as a model-fitness signal rather than a reward trace.

Second, the tracked-only ablation separated deployment from inference. There, β_k was updated but could not modulate policy precision, and behavior remained close to the no-affect baseline. Full partner-local precision lowered policy entropy through the $\beta_k \to \gamma_k$ pathway. The mechanism therefore changed how strongly the current partner model was expressed in action. Behavior changed even without a corresponding improvement in partner-state inference.

Third, the shared-β comparison showed why confidence should remain relationship-local. Pooling evidence across partners preserved a coarse reliability signal but weakened the partner-specific dissociation between surprise and payoff. In multi-partner settings, evidence from one relationship need not change confidence in another. Partner-local β_k keeps confidence tied to the partner whose behavior generated it. This complements active-inference accounts of theory of mind, which model what another agent knows or intends [17], and factorized active-inference approaches that separate latent components of strategic interaction [18]. Partner-local affective precision instead addresses the confidence placed in the current partner model during action selection.

The betrayal and mixed-volatility experiments show when accumulated confidence becomes fragile. After a previously cooperative partner changes behavior, confidence built under the earlier model can become stale. In the abrupt-betrayal condition, partner-local precision lowered policy entropy and increased joint partner-type accuracy, while the payoff advantage remained small and uncertain. The consistent effect is deployment change under social volatility. Its payoff consequences depend on the speed of change, the availability of alternative partners, and how quickly engagement can be reallocated. A volatility or change-detection signal could discount accumulated confidence when prediction-error structure changes, similar in spirit to hierarchical volatility models [15, 3] but operating at the level of action confidence rather than learning rate.

The profile analyses varied precision gain α and prior model fitness. Low-gain profiles updated weakly and distributed engagement diffusely, whereas high-gain profiles showed larger confidence swings and stronger reactivity. In the mixed-volatility setting, stronger gain increased payoff while also increasing false-positive confidence reduction toward stable partners. Social trust calibration therefore separated into update speed, confidence amplitude, reengagement after repair, sensitivity to change, and vulnerability to false positives, rather than varying along a single high-versus-low trust axis. These profiles connect naturally to empirical work on social learning, psychosis, and attachment [3, 1, 5], but they have not yet been fitted to human behavior. They provide predictions for future behavioral fitting.

The current implementation leaves several directions for future work. First, β_k is updated outside the agent's generative model. Embedding β_k as a hidden state would allow the agent to represent expected changes in its own confidence and choose policies that anticipate changes in partner reliability [13]. Second, partners are fixed environment policies rather than active-inference agents. This isolates the focal mechanism; reciprocal belief and confidence updating remain for future work. Third, the profile analyses generate behavioral predictions rather than fitted individual-difference estimates. Testing whether α and prior model-fitness parameters capture human trust variation will require fitting the model to behavioral time series. Finally, some effects are small at the current simulation scale, especially the betrayal payoff difference and the 20-seed profile analyses. The strongest empirical support comes from the dissociations that remain stable across controls: predictability over value, deployment over inference, and partner-local over shared confidence.

Two next steps follow directly. The first is empirical fitting to human behavior. The model produces behavioral quantities that can be measured from trust-game time series, including exploitation sensitivity, betrayal recovery, reengagement after repair, false-positive confidence reduction, and partner reallocation under mixed volatility. Estimating α and prior model-fitness parameters from individual participants would test whether these computational profiles correspond to stable differences in social trust calibration. The second is reciprocal multi-agent modeling, replacing environment-side partners with active-inference agents that maintain their own beliefs, policies, and precision estimates. This would allow partner-local affective precision to be studied together with theory-of-mind inference in richer multi-agent systems.

5 Conclusion

Predicting what a partner will do, modeling what they know or intend, and deciding how confidently to act on one's current partner model are separable problems. Partner-local affective precision addresses the third as a relationship-specific metacognitive confidence signal. The simulations show that it tracks predictive reliability more than realized value, affects action deployment through policy precision rather than improved partner-state inference, and creates temporally structured trust-revision profiles under social change. By giving each social relationship its own model-fitness estimate, the mechanism allows action confidence to be calibrated independently across a social portfolio. Precision gain and priors over model fitness define a family of computational trust profiles for individual-difference research, providing a computational account of how social affect can regulate confidence in action.

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Appendix

A Full POMDP Specification

This appendix specifies the POMDP components used by the focal agent.

A.1 Hidden state factors and observations

The focal agent's generative model for each partner k maintains three latent factors:

$$\begin{aligned} s_k^{\text{type}} &\in \{\text{cooperator, reciprocator,} \\ &\qquad \text{exploiter, random}\} && \text{(latent behavioral type)} \quad (4) \\ s_k^{\text{stance}} &\in \{\text{trusting, neutral, hostile}\} && \text{(latent stance)} \quad (5) \\ s^{\text{own}} &\in \mathcal{A} && \text{(agent's executed social action)} \quad (6) \end{aligned}$$

The own-action state is *deterministic bookkeeping*: the agent always knows its own action, and this factor makes that knowledge available to the observation model. It is not a psychological hidden variable. In the binary task, the joint hidden state per partner spans $4 \times 3 \times 2 = 24$ possibilities; graded regimes replace the binary action factor with the configured investment levels.

Observations are partner action and payoff:

$$\begin{aligned} o_k^{\text{act}} &\in \{\text{cooperate, defect}\} && \text{(partner response)} \quad (7) \\ o_k^{\text{pay}} &\in \{-1, 1, 3, 5\} && \text{(focal agent's realized payoff)} \quad (8) \end{aligned}$$

A.2 Partner-action likelihood

The core of the generative model is the *partner cooperation likelihood table*, which defines the probability the partner cooperates given type and stance:

Table 3: Cooperation likelihood $P(\text{cooperate} \mid s^{\text{type}}, s^{\text{stance}})$.

Partner type	Trusting	Neutral	Hostile
Cooperator	0.95	0.80	0.55
Reciprocator	0.90	0.70	0.30
Exploiter	0.70	0.35	0.10
Random	0.60	0.50	0.35

This table is the foundation of the likelihood modality A_k^{act}. Entries define

$$P\left(o_k^{\text{act}} = \text{cooperate} \mid s_k^{\text{type}}, s_k^{\text{stance}}\right), \quad (9)$$

with $P(o_k^{\text{act}} = \text{defect} \mid \cdot) = 1 - P(o_k^{\text{act}} = \text{cooperate} \mid \cdot)$. The modality is uniform over s^{own}: partner action depends only on inferred type and stance. This operationalizes what the agent considers a predictable partner—a cooperative reciprocator in a trusting stance has high predicted cooperation; an exploiter in a hostile stance has low predicted cooperation.

A.3 Payoff likelihood

Payoff enters as a second observation modality $A_k^{\text{pay}} = P(o_k^{\text{pay}} \mid s^{\text{own}}, s_k^{\text{type}}, s_k^{\text{stance}})$ rather than as external reward. The environment payoff table is deterministic (Table 1 in the main text); in the generative model, partner action is marginalized through the cooperation likelihood:

$$P\big(o_k^{\text{pay}} = v \mid s^{\text{own}}, s_k^{\text{type}}, s_k^{\text{stance}}\big) = \sum_{a^{\text{partner}} \in \{\text{C},\text{D}\}} \mathbb{I}\big[\pi(s^{\text{own}}, a^{\text{partner}}) = v\big]\, P\big(o_k^{\text{act}} = a^{\text{partner}} \mid s_k^{\text{type}}, s_k^{\text{stance}}\big)\,, \tag{10}$$

where $\pi(\cdot,\cdot)$ maps joint actions to focal payoffs $\{-1, 1, 3, 5\}$ and $P(o_k^{\text{act}} \mid \cdot)$ comes from Eq. (9). This preserves the Prisoner's Dilemma incentive structure at horizon 1 while allowing trust-building dynamics to emerge over longer horizons.

A.4 Transition dynamics

Type drift (B_k^{type}). Partner type is uncontrollable and drifts slowly:

$$P(s_k^{\text{type}\prime} = j \mid s_k^{\text{type}} = i) = \begin{cases} 1 - p_{\text{switch}} & j = i, \\ p_{\text{switch}}/(K_{\text{type}} - 1) & j \neq i, \end{cases} \tag{11}$$

with $K_{\text{type}} = 4$ and default $p_{\text{switch}} = 0.05$. Scheduled betrayal and mixed-volatility scenarios set $p_{\text{switch}} = 0$.

Stance dynamics (B_k^{stance}). Stance transitions are action-dependent on the focal agent's social action a:

$$P\big(s_k^{\text{stance}\prime} \mid s_k^{\text{stance}}, a\big)\,, \tag{12}$$

as specified below. Cooperation tends to build trust; defection damages trust; recovery from hostility is slower than collapse.

Table 4: Stance transition dynamics $P(s'^{\text{stance}} \mid s^{\text{stance}}, a)$ conditional on focal agent action.

Agent cooperates			
To \ From	Trusting	Neutral	Hostile
Trusting	0.90	0.30	0.05
Neutral	0.10	0.60	0.35
Hostile	0.00	0.10	0.60
Agent defects			
To \ From	Trusting	Neutral	Hostile
Trusting	0.10	0.05	0.02
Neutral	0.50	0.35	0.18
Hostile	0.40	0.60	0.80

These dynamics instantiate the asymmetry of trust: cooperation gradually builds trust (neutral $\to$ trusting in $\sim$ 3–4 steps), while defection rapidly damages trust (trusting $\to$ hostile in $\sim$ 2 steps); recovery from hostility is slow ($\sim$ 4–5 steps to neutral).

Own-action bookkeeping (B^{own}). The own-action factor updates deterministically from the executed social action:

$$P(s^{\text{own}\prime} = a' \mid s^{\text{own}}, a) = \mathbb{I}[a' = a]. \tag{13}$$

This factor is not a psychological hidden variable; it makes the agent's executed action available to the payoff likelihood in standard factorized form.

A.5 Preferences, priors, and policy priors

Observation preferences (C_k). There is no direct preference over partner actions:

$$C_k^{\text{act}} = [0,\, 0] \quad \text{over } \{\text{cooperate}, \text{defect}\}. \tag{14}$$

Payoff preferences ascend over $\{-1, 1, 3, 5\}$:

$$C_k^{\text{pay}} = \text{log\_softmax}\left(\frac{[-1, 1, 3, 5]}{\tau}\right), \tag{15}$$

where τ is a temperature parameter controlling preference sharpness. Instrumental motivation therefore enters through payoff observations and multi-step planning rather than direct action preferences.

Initial-state priors (D_k). Type, stance, and own-action priors are

$$D_k^{\text{type}} = \left[\tfrac{1}{4}, \tfrac{1}{4}, \tfrac{1}{4}, \tfrac{1}{4}\right], \tag{16}$$

$$D_k^{\text{stance}} = [0.2,\, 0.6,\, 0.2], \tag{17}$$

$$D^{\text{own}} = [0.5,\, 0.5] \quad \text{(binary task; uniform over graded levels in graded regimes).} \tag{18}$$

Policy priors (E_k). Policy priors are uniform over admissible action sequences at the configured planning horizon (no habitual bias toward cooperation or defection). For horizon 1 in the binary task, $E_k = [0.5, 0.5]$ over $\{\text{cooperate}, \text{defect}\}$; longer horizons enumerate all length-τ sequences with equal prior mass.

Partner selection across k is handled outside the per-partner POMDP by evaluating policy branches for each partner with precision-adjusted logits before executing the chosen action (Appendix B).

A.6 Partner implementation

Reported experiments use one focal active-inference agent with per-partner generative models as described above. Partners are parameterized environment-side policies, not reciprocal active-inference agents. Each partner maintains a latent type and stance, follows the type-by-stance cooperation probabilities in Table 3, and undergoes action-dependent stance transitions in Table 4. Partners do not run variational inference, maintain their own policy precision γ_k, or update affective estimates β_k. This design isolates the focal agent's precision-tracking mechanism before adding reciprocal belief and policy updating on the partner side.

B Affective Precision Update and Policy Selection

This appendix gives the β-update rule, policy-precision mapping, and partner-choice policy-selection procedure.

B.1 Categorical beta support

Each partner maintains a categorical posterior $q(\beta_k)$ over discrete levels $\{0.5, 0.67, 1.0, 1.5, 2.0\}$, updated each round from affective charge ϕ_k. The posterior is categorical rather than continuous; this keeps the inverse-β convention identifiable while preserving a compact implementation of the precision dynamics.

B.2 Persistence prior

The update proceeds in three steps. First, a persistence prior smooths the previous posterior via a tridiagonal transition matrix:

$$p(\beta_k) \leftarrow T \cdot q_{\text{prev}}(\beta_k), \tag{19}$$

where T is tridiagonal with persistence parameter 0.8 and reflecting boundaries, encoding the prior expectation that precision evolves slowly.

B.3 Charge-based pseudo-likelihood

Second, a pseudo-likelihood is constructed directly from affective charge, mapping positive charge to high precision (low β) and negative charge to low precision (high β):

$$\log \ell(\beta_\ell) = \phi_k \cdot \frac{1}{\beta_\ell}, \tag{20}$$

where β_ℓ ranges over the five support points. This is a hand-designed update rule (not derived from the POMDP generative model) chosen to satisfy monotonicity: strong positive charge reinforces low β, strong negative charge reinforces high β, and the effect scales with effective precision at each level.

B.4 Posterior update and policy precision

Third, the posterior is normalized:

$$q(\beta_k) \propto \exp\left(\log \ell(\beta_\ell)\right) \cdot p(\beta_k). \tag{21}$$

The agent then uses the posterior mean to set policy precision:

$$\gamma_k = \gamma_{\text{base}} \,/\, \mathbb{E}_{q(\beta_k)}[\beta_k]. \tag{22}$$

The β_k posterior is maintained as an external auxiliary tracker rather than a proper hidden state in the generative model, which simplifies implementation at the cost of losing the agent's ability to form prior expectations over its own future affective states (discussed in Section 4).

B.5 Centered cross-partner policy logits

In partner-choice regimes, precision must sharpen policies within a partner branch without creating artifacts when branches are compared across partners. Let $u_{k,\pi}$ denote the pre-precision policy logit for policy π toward partner k, and let $\bar{u}_k$ be that partner branch's mean logit. Precision is applied to centered logits:

$$\tilde{u}_{k,\pi} = \bar{u}_k + \gamma_k \left(u_{k,\pi} - \bar{u}_k\right). \tag{23}$$

The agent then samples or selects policies from the combined set of $\tilde{u}_{k,\pi}$ values. Centering preserves the mean score of each partner branch while allowing γ_k to sharpen or flatten the within-partner policy distribution. This prevents low-precision scaling from moving a uniformly poor branch toward zero and making it spuriously attractive in cross-partner comparison.

B.6 Ablation variants

Reported comparisons use four affect-runtime variants. The *no-affect* control disables the beta tracker and fixes policy precision at γ_{base}. The *tracked-only* ablation updates $q(\beta_k)$ normally but fixes $\gamma_k = \gamma_{\text{base}}$, allowing the precision estimate to be analyzed without letting it affect action selection. The default *partner-local precision* variant applies the full update and sets γ_k from each partner's posterior mean. The *shared-β* ablation maintains a single pooled beta state updated from partner evidence while keeping partner-local POMDP beliefs intact; all partner branches then inherit precision from the shared expected β.

C Simulation Protocols and Metrics

This appendix summarizes simulation conditions, replication counts, schedules, and metrics.

C.1 Simulation conditions

All reported experiments use one focal active-inference agent with four partners ($K = 4$), per-partner generative models, and environment-side partner policies (Appendix A). Conditions differ by payoff regime, assignment mode, and scheduled partner changes.

Binary random-assignment baseline. In the binary trust game (Table 1 in the main text), partners are assigned randomly each round and the agent chooses cooperate or defect. Stochastic partner-type drift uses $p_{\text{switch}} = 0.05$ unless disabled for scheduled-switch scenarios.

Open graded decision regime. The graded regime replaces the binary action with a discretized investment level while keeping the same latent type–stance structure. This is the primary deployment-dissociation setting (Section 3.3): affect, no-affect, and tracked-only variants are compared on policy entropy and cumulative payoff.

Partner-choice regime. The agent first selects a partner, then selects an action toward that partner each round. Partner selection is where engagement reorganization and social-allocation readouts are defined (Section 3.4).

Focal-switch locality probe. A partner-choice graded environment with type volatility disabled ($p_{\text{switch}} = 0$) and a single scheduled stance switch on partner P0 tests whether local versus shared β tracking preserves the surprise-over-reward dissociation (Section 3.1).

Abrupt betrayal. A partner-choice graded environment with scheduled stance betrayal on a previously cooperative partner tests post-switch reallocation, entropy, payoff, and joint accuracy (Section 3.5).

Precision-dynamics perturbations. Computational profile variants (low-gain, high-gain, and cautious-prior profiles) perturb β_k amplitude and stability in the graded decision regime (Appendix D).

Extended profile experiments (Exp A–D). Parameter sweeps and factorial designs vary precision gain α and prior over model fitness to define computational trust-calibration profiles in open graded, betrayal, partner-choice, forgiveness, and mixed-volatility environments (Section 3.6).

C.2 Seed counts and episode lengths

Unless otherwise stated in the main text, episode length is 200 rounds. The abrupt-betrayal condition uses 120 rounds because post-switch readouts dominate interpretation. Profile-scale and mixed-volatility tables use 20 seeds per condition. Central behavioral comparisons use 30 seeds per condition. Exploratory diagnostic runs used three seeds per variant and were used only for implementation validation; all main-text claims are based on the confirmation-scale or profile-scale runs reported in the relevant sections.

Representative configured scales:

Table 5: Representative episode lengths and replication counts by experiment family.

Experiment family	Setting	Rounds	Seeds
H0/H2 open graded deployment	graded, random or open choice	200	30
H1 model fitness	graded / reliability probes	200	30
H3 focal-switch locality	partner choice, single switch	100	30
H4 partner choice	partner choice	200	30
H5 abrupt betrayal	partner choice, scheduled betrayal	120	30
H6 perturbation dynamics	graded perturbations	200	30
Exp A–D profiles	sweeps / factorial / forgiveness / mixed volatility	200	20

Supplementary materials. The supplementary repository contains the simulation environments, affect-runtime variants, experiment configuration files, analysis scripts, and figure-generation code used for the reported experiments.

C.3 Betrayal schedule

The canonical abrupt-betrayal protocol (H5 `betrayal_choice`) uses graded payoffs, agent-chosen partners, four partners with fixed initial types and stances, and no stochastic type switching ($p_{\text{switch}} = 0$). Partner P0 begins as a cooperator in a trusting stance. At round 31, P0 undergoes a scheduled stance switch to hostile while other partners retain their initial configurations. Primary readouts are post-switch cumulative payoff, policy entropy, partner reallocation, and joint partner-type accuracy over the remaining episode.

The profile-scale betrayal environment used in Exp A and Exp B applies the same single-switch structure within 200-round episodes for recovery-time and quadrant metrics. Exp C (forgiveness) extends this schedule: rounds 1–80 cooperative baseline, rounds 81–120 betrayal on P0, rounds 121–200 reversion to cooperative/trusting on P0.

C.4 Mixed-volatility schedule

The mixed-volatility environment (Exp D; Section 3.6) places four partners with heterogeneous volatility in the same partner-choice episode over 200 rounds:

- **P0:** stationary cooperator (no scheduled change).
- **P1:** stationary exploiter (no scheduled change).
- **P2:** abrupt shift—cooperative/trusting pre-switch, hostile/exploiter from the scheduled switch onward (round 101 in the implemented scenario).
- **P3:** gradual drift—stance and type transitions spread across mid-episode rounds (neutral stance from round 51; hostile/exploiter shift from round 151 in the implemented scenario).

Stochastic type switching is disabled ($p_{\text{switch}} = 0$). Conditions compare default, low-α, high-α, and no-affect variants at 20 seeds each.

C.5 Partner-choice allocation metrics

Partner-choice readouts quantify how precision reshapes *who* the agent engages, not only how it acts toward a fixed partner.

Selection share. Per-partner selection rate is the fraction of rounds on which each partner is chosen. Partner-choice reallocation is summarized qualitatively in Section 3.4.

Selection concentration. The Gini coefficient of the partner-selection distribution summarizes how concentrated engagement is across the social portfolio (used in profile Exp A and Exp B).

Post-betrayal allocation. Post-switch windows report reallocation away from the switched partner and toward alternatives, including post-betrayal P0 selection and high-investment rates in profile analyses.

Policy entropy under choice. Mean q_π entropy in partner-choice regimes captures how sharply the agent commits to partner–action policies (reported comparatively for affect versus no-affect controls).

C.6 Entropy, payoff, and accuracy readouts

Total payoff. Cumulative focal-agent payoff over the episode (or over defined post-switch windows for betrayal analyses).

Policy entropy. Mean q_π entropy (‘q_pi_entropy‘ in logged outputs) measures action-commitment sharpness; lower entropy indicates sharper policy selection under precision modulation.

Joint partner-type accuracy. Mean joint correctness of inferred partner type (‘mean_joint_accuracy‘) summarizes epistemic state inference; it is reported separately from payoff because precision modulation can change deployment and accuracy independently of cumulative reward.

β_k **range.** Per-partner or episode-aggregated range of β_k support summarizes precision-dynamics amplitude in profile and perturbation tables.

Model-fitness dissociation. Partner-local correlations between mean precision and action surprisal versus realized payoff test whether β_k tracks predictive reliability rather than partner value (Section 3.1).

Mixed-volatility discrimination metrics. The discrimination index is the Pearson correlation between per-partner β_k trajectory and per-partner behavioral stability (higher indicates better separation of stable versus shifting partners). The P0 false-positive rate is the rolling rate at which engagement with the stationary cooperator P0 falls more than 15% below its early baseline despite P0 never changing.

Betrayal timing metrics. Recovery time and related latency readouts measure rounds after a switch until selection or payoff returns toward baseline thresholds (profile Exp A/B and betrayal analyses).

C.7 Replication summary

Central behavioral comparisons in Section 3 use 30 seeds per condition. Profile-scale and mixed-volatility analyses use 20 seeds per condition. The model-fitness dissociation (Section 3.1) uses active-encounter partial correlations to compare precision–surprise against precision–payoff associations, with shared β included as the relational-specificity ablation.

D Extended Profile and Robustness Results

This appendix reports extended parameter sweeps, profile analyses, and robustness diagnostics that support the main-text results.

D.1 Precision-dynamics perturbations

We first perturb the amplitude and stability of β_k dynamics to test how precision dynamics affect action commitment. These variants are computational controls rather than diagnostic categories.

Table 6: Precision-dynamics perturbations in the graded decision regime.

Variant	β_k range	Total payoff	Policy entropy
affect	1.32	1851.3	8.6
low-gain	0.24	1840.5	8.9
high-gain	1.48	1859.0	8.1
cautious-prior	1.48	1877.7	8.4

Flattening the precision signal produces the clearest behavioral change. The low-gain variant compresses β_k range to 0.24 and raises entropy to 8.9, near the no-affect and tracked-only baselines. Expanded-dynamics variants produce wider

β_k movement, but payoff does not follow range monotonically. The parameter sweeps below therefore vary gain and prior structure directly.

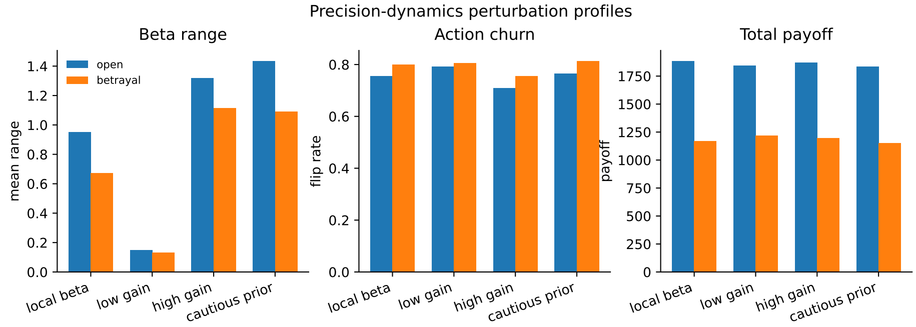

Fig. 5: Precision-dynamics perturbation profiles in the graded decision regime. Variants compare precision-dynamics settings in the graded decision regime.

D.2 Alpha sweep

Sweeping α across $\{0.05, 0.1, 0.3, 0.5, 1.0, 2.0, 4.0, 8.0\}$ tests whether precision gain controls reactive social confidence across metrics. β_k movement range increases monotonically with α in the betrayal regime:

Table 7: β_k range by precision-gain α (betrayal regime, 20 seeds).

α	Mean β_k range
0.05	0.147
0.1	0.182
0.3	0.326
0.5	0.416
1.0	0.580
2.0	0.862
4.0	1.051
8.0	1.218

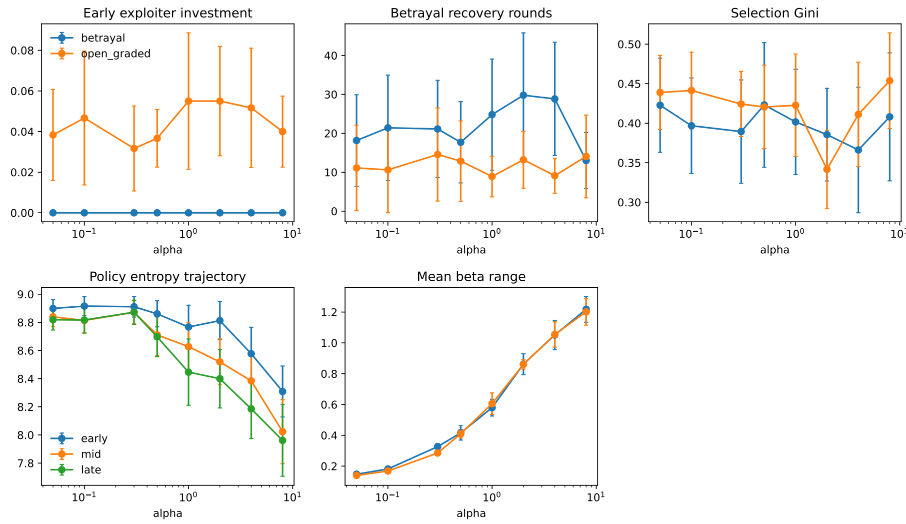

Fig. 6: Precision-gain sweep across open graded and betrayal environments (20 seeds). Higher α expands β_k range and reshapes engagement readouts without monotonic payoff ranking.

D.3 Profile quadrant profiles

Crossing prior belief (naive versus cautious) with α (low versus high) defines four behavioral profiles tested in the betrayal environment (20 seeds). Table 8 gives key metrics.

Table 8: Behavioral metrics by profile (betrayal environment, 20 seeds). Default agent shown for reference.

Profile	Payoff	Recovery	Gini	β range	Trust asymmetry
Anxious-reactive	1932	26.5	0.338	1.084	0.98
Hypervigilant	1951	20.1	0.423	0.892	1.65
Naive-stubborn	1894	21.3	0.304	0.364	1.19
Avoidant-rigid	1889	14.9	0.357	0.322	1.21
Default	1991	23.0	0.423	0.893	1.68

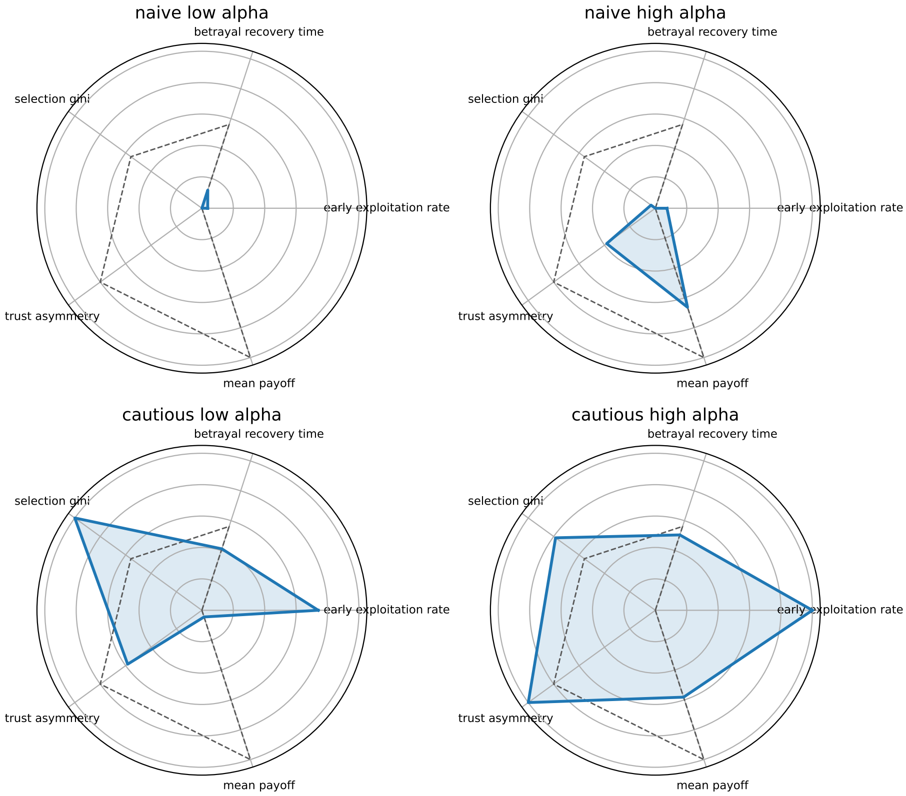

Fig. 7: Gain-prior profile quadrants from the prior-by-α factorial (betrayal environment, 20 seeds).

D.4 Forgiveness and trust repair

The forgiveness paradigm uses a 200-round partner-choice episode: cooperative baseline (rounds 1–80), betrayal on P0 (rounds 81–120), then reversion to cooperative/trusting behavior (rounds 121–200). Readouts include reengagement latency, post-return selection, payoff recovery, and β_k recovery trajectories for the reverted partner.

Table 9: Forgiveness and trust-repair metrics by precision profile (20 seeds). Reengagement is the post-repair P0 selection rate; payoff recovery compares late post-repair payoff with the late pre-betrayal baseline.

Variant	Reengagement	Payoff recovery	Latency	β_k range
cautious-high-α	0.518	0.996	5.5	0.971
cautious-low-α	0.630	1.014	1.5	0.287
default	0.475	1.019	3.0	1.012
naive-high-α	0.560	1.005	2.2	1.100
naive-low-α	0.527	1.004	4.6	0.325
no-affect	0.593	1.033	1.7	–

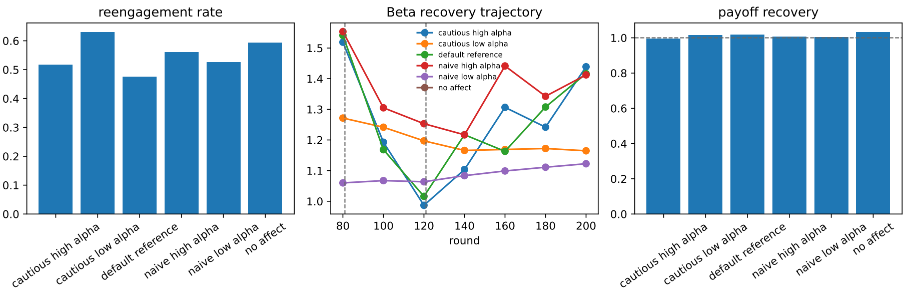

Fig. 8: Forgiveness/trust-repair diagnostic from the 20-seed profile analyses. Reengagement and payoff recovery do not collapse into a single monotonic gain effect: no-affect and cautious-low-α profiles reengage most often, while payoff recovery remains near baseline across profiles.

D.5 Mixed-volatility extended diagnostics

The primary mixed-volatility table and figure remain in Section 3.6. Extended Exp D diagnostics include per-partner β_k trajectories, rolling P0 selection concentration, and pairwise stability–precision correlations used to compute the discrimination index and P0 false-positive rate reported in the main text.